

EggHand: A Multimodal Foundation Model for Egocentric Hand Pose Forecasting

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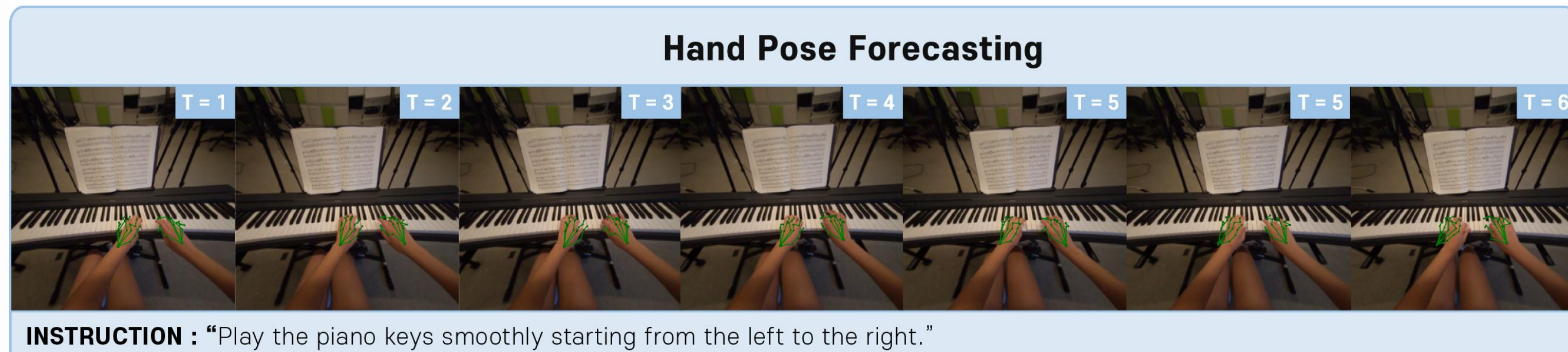
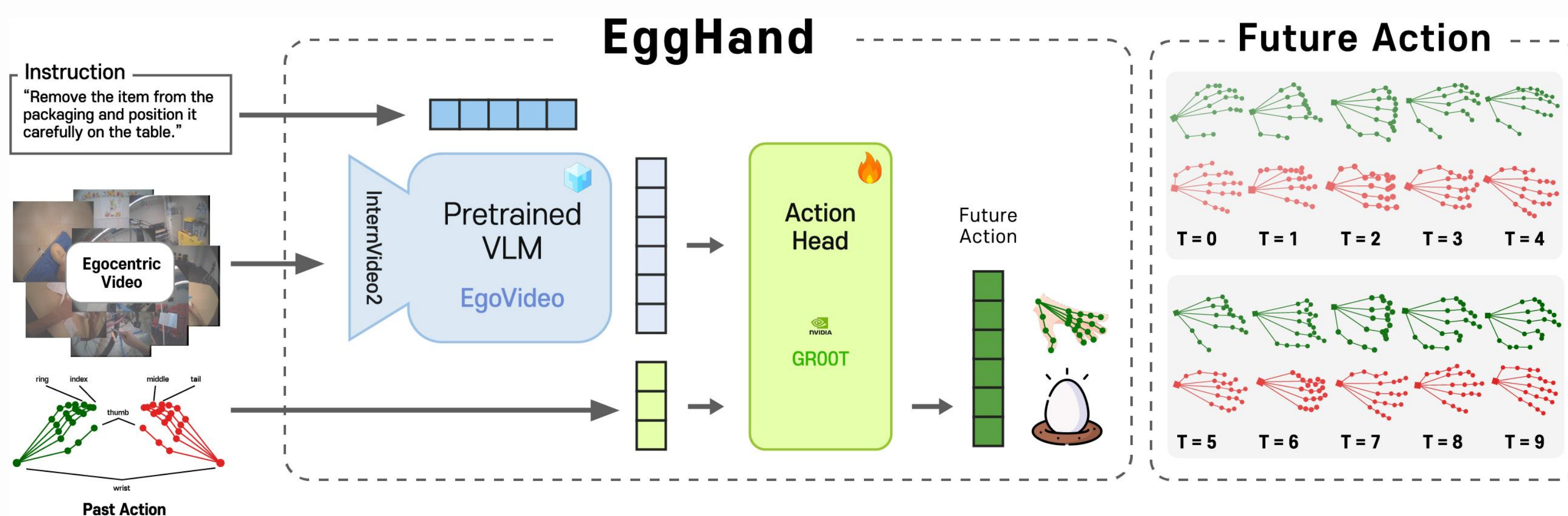


01 Introduction

- ✓ Egocentric 3D hand pose forecasting is essential for anticipating human intent in AR/VR and human-robot interaction.
- ✓ However, existing methods rely on low-level motion cues, which fail to capture high-level semantics and degrade under severe camera ego-motion.
- ✓ We propose EggHand, combining a Vision-Language-Action (VLA) action decoder with an egocentric video encoder for robust, viewpoint-aware hand pose forecasting.

02 Motivation

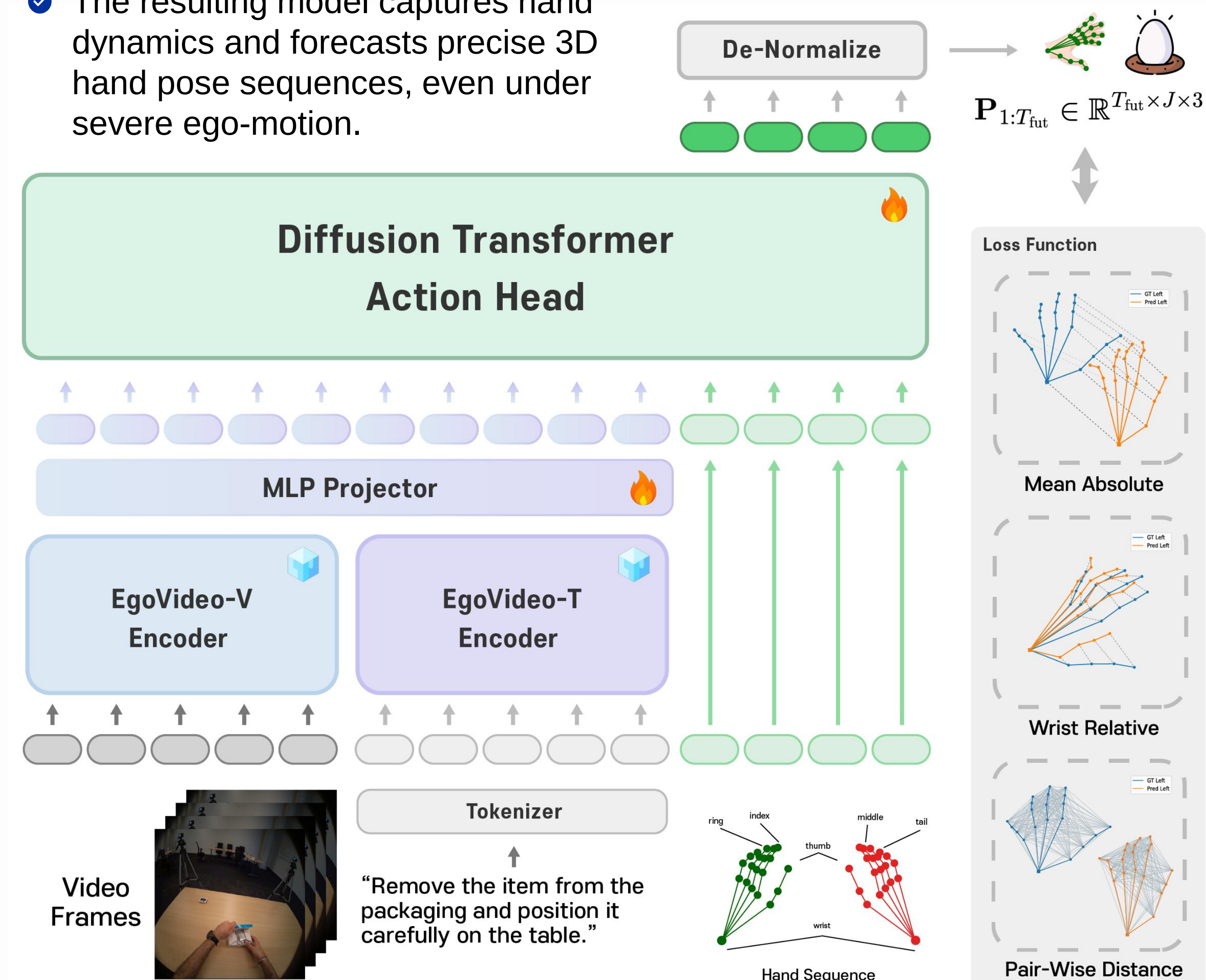
- ✓ While VLA models excel at generating physically grounded actions, they have not yet been applied to egocentric 3D hand pose forecasting.
- ✓ However, VLM encoders fall short on fine-grained hand kinematics and joint constraints under severe camera motion.



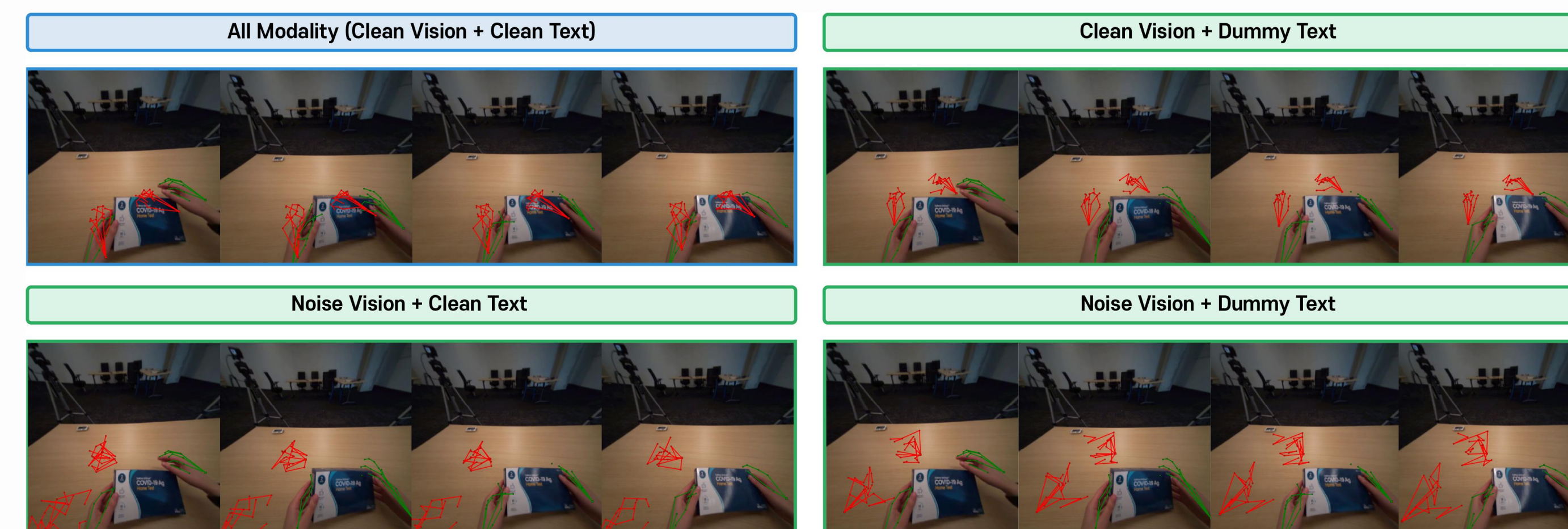
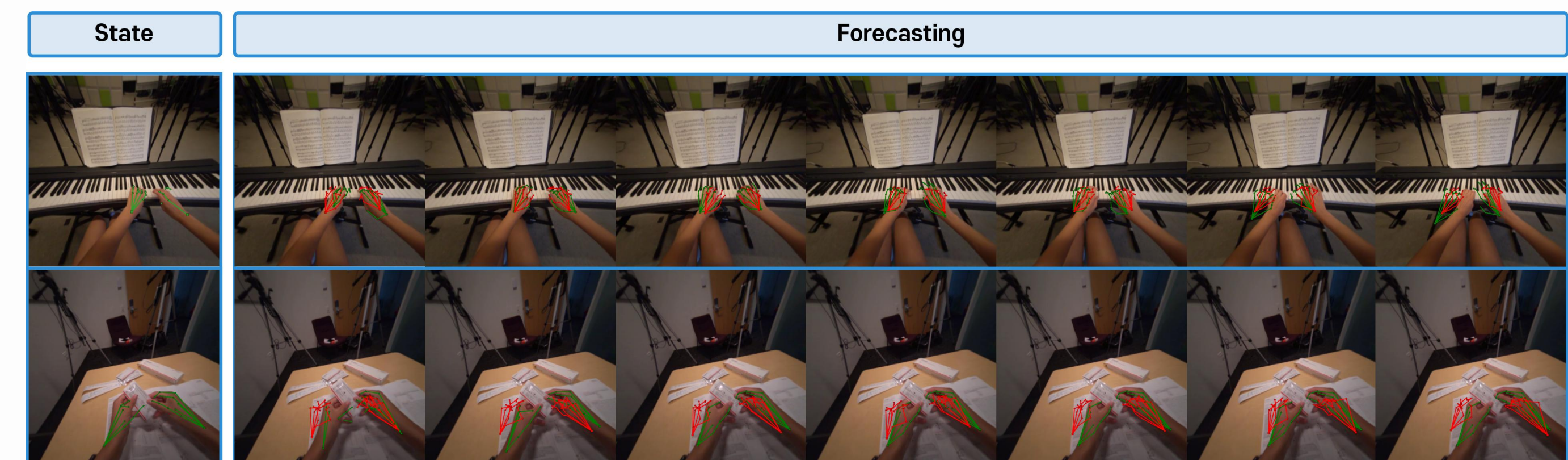
03 Methodology

Key Idea

- ✓ **Foundation Model Domain Adaptation:** Fusing an egocentric foundation encoder with a VLA decoder for stable, kinematics-aware forecasting.
- ✓ We bridge a frozen egocentric encoder (EgoVideo) and a GR00T-based VLA decoder with lightweight adapters, blending spatial context with temporal action priors.
- ✓ The network is trained with a geometry-aware loss that jointly enforces absolute accuracy, wrist-relative stability, and intra-hand joint consistency.
- ✓ The resulting model captures hand dynamics and forecasts precise 3D hand pose sequences, even under severe ego-motion.



04 Results



Method	Trajectory		Pose	
	ADE	FDE	MPJPE	MPJPE-F
Static	0.335	0.405	0.166	0.179
CVM	0.346	0.467	0.166	0.183
USST	0.562	0.581	-	-
EgoH4	0.267	0.333	0.116	0.141
Ours	0.271	0.271	0.076	0.077

Method	Top 10%		All	
	ADE	MPJPE	ADE	MPJPE
EgoH4	0.294	0.139	0.267	0.116
Ours w. InternVL3.5-1B	0.303	0.112	0.287	0.082
Ours	0.276	0.106	0.271	0.076

05 Conclusion

- ✓ We presented EggHand, the first framework to harness VLA models for robust, semantics-aware egocentric 3D hand pose forecasting.
- ✓ EggHand achieves state-of-the-art performance on EgoExo4D, bridging foundation models and fine-grained embodied AI.